

Code:

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#include <stdio.h>
#include <stdlib.h>
#include <stdbool.h>

#define MAXN 100005

int boy_crush[MAXN];
int girl_crush[MAXN];
int beaten[MAXN];
int target_of[MAXN];

int main() {
    int T;
    if (scanf("%d", &T) != 1) return 0;

    while (true) {
        if (T-- <= 0) break;

        int N;
        if (scanf("%d", &N) != 1) break;

        for (int i = 1; i <= N; i++) {
            scanf("%d", &boy_crush[i]);
            beaten[i] = 0;
        }
        for (int i = 1; i <= N; i++) {
            scanf("%d", &girl_crush[i]);
        }

        // Boy i crushes on girl_crush, who in turn crushes on target_of[i]
        for (int i = 1; i <= N; i++) {
            int g = boy_crush[i];
            int z = girl_crush[g];
            target_of[i] = z;
            if (i != z) {
                beaten[z]++;
            }
        }

        int max_beatings = 0;
```

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for (int i = 1; i <= N; i++) {
    if (beaten[i] > max_beatings) {
        max_beatings = beaten[i];
    }
}

// Mutual beatings: pair (i, j) with i < j where target_of[i] == j and target_of[j] == i
int mutual_pairs = 0;
for (int i = 1; i <= N; i++) {
    int j = target_of[i];
    if (j > i && target_of[j] == i) {
        mutual_pairs++;
    }
}

printf("%d %d\n", max_beatings, mutual_pairs);
}

return 0;
}

```

Input:

```

2
3
2 2 1
3 2 1
4
2 3 4 1
2 3 4 1

```

Output:

1 0

1 4